

Transport Layer



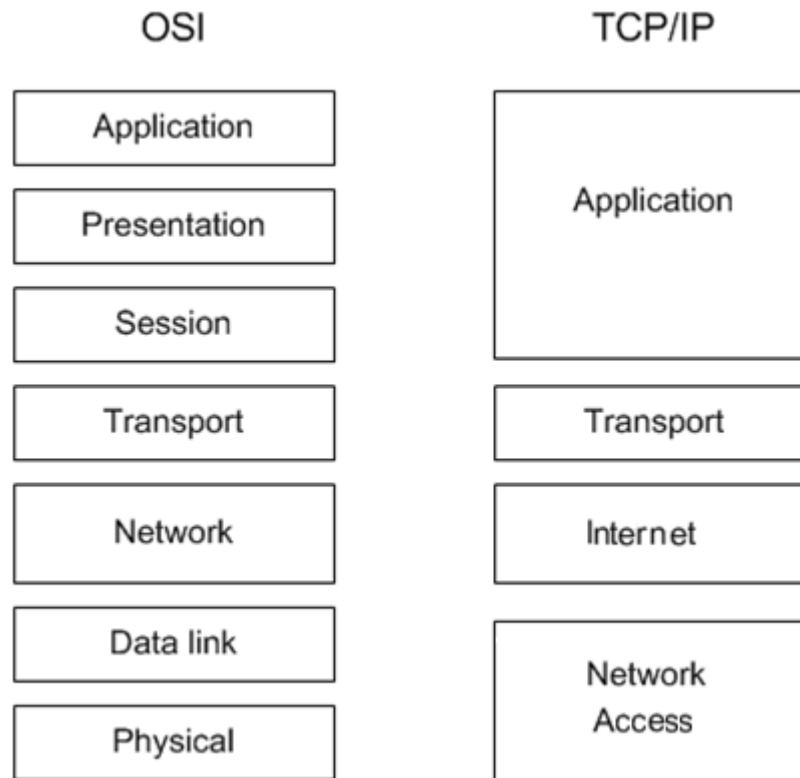
CCNA 1 ITN - Chapter 9

Objectives

- Describe the functions of L4: Transport
- Analyze different services: Voice, Mail.
- Describe TCP - Describe UDP
- Port Numbers
- TCP Header – Deep Analyze
- The Start/End of the connection
- UDP Header – Deep Analyze

Transport Layer Functions

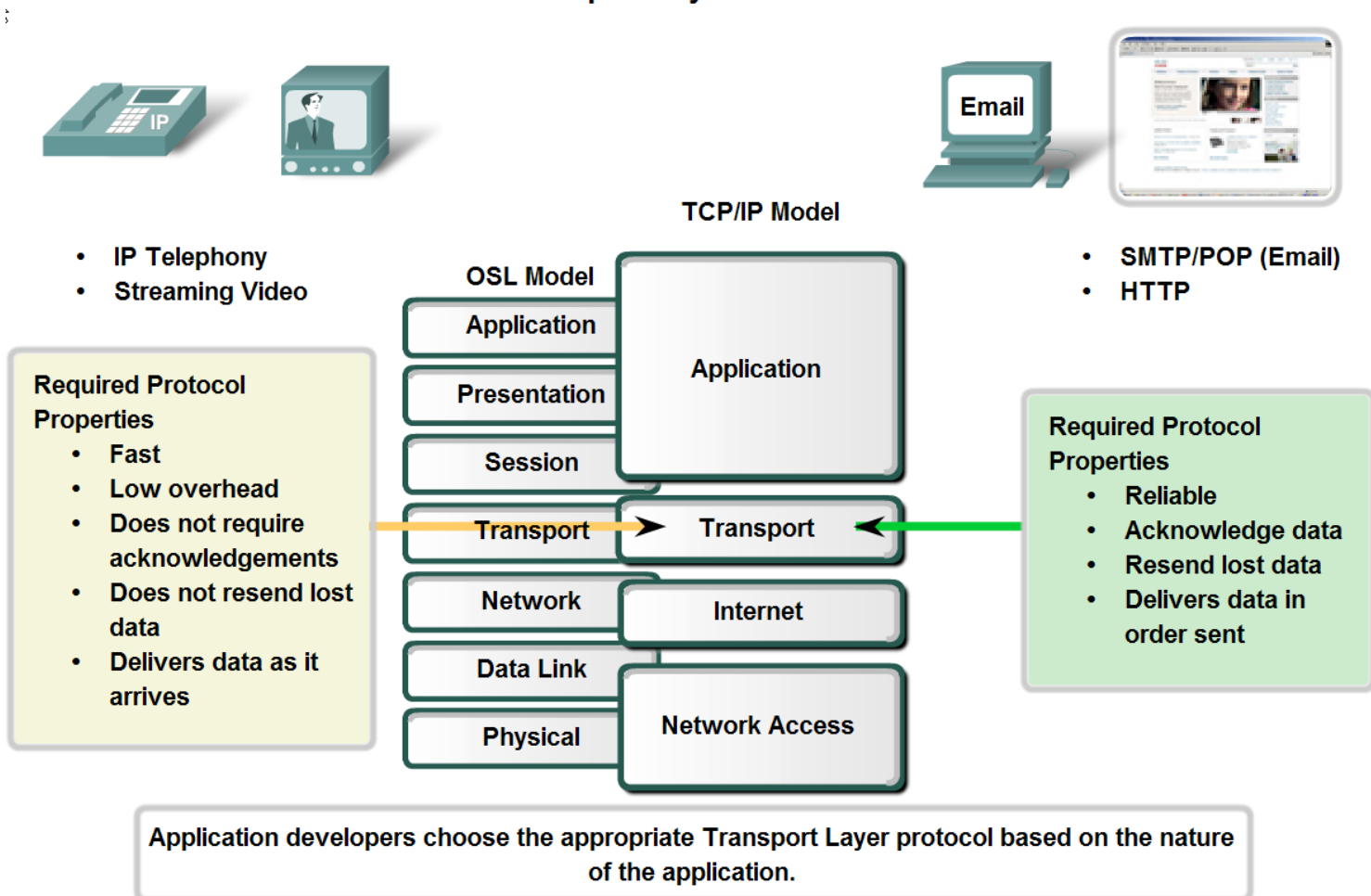
- ◆ Service Management
- ◆ Segmentation and Reassembly
- ◆ Transmissions and Retransmissions
- ◆ PDU Layer Transport = Segment



Transport Layer Functions

- ◆ Service Management: Different Services = Different Needs

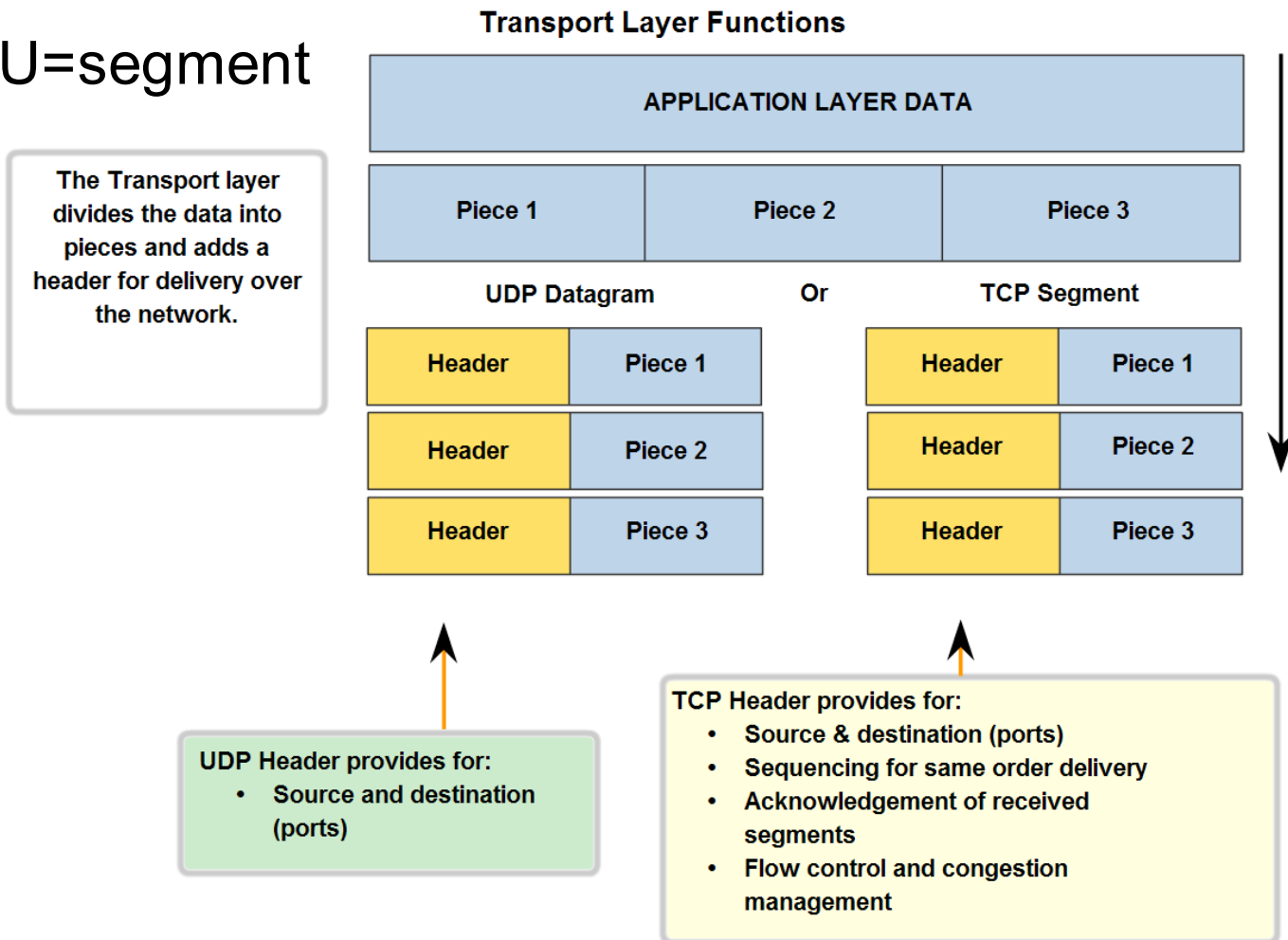
Transport Layer Protocols



Transport Layer Functions

◆ Segmentation and Reassembly

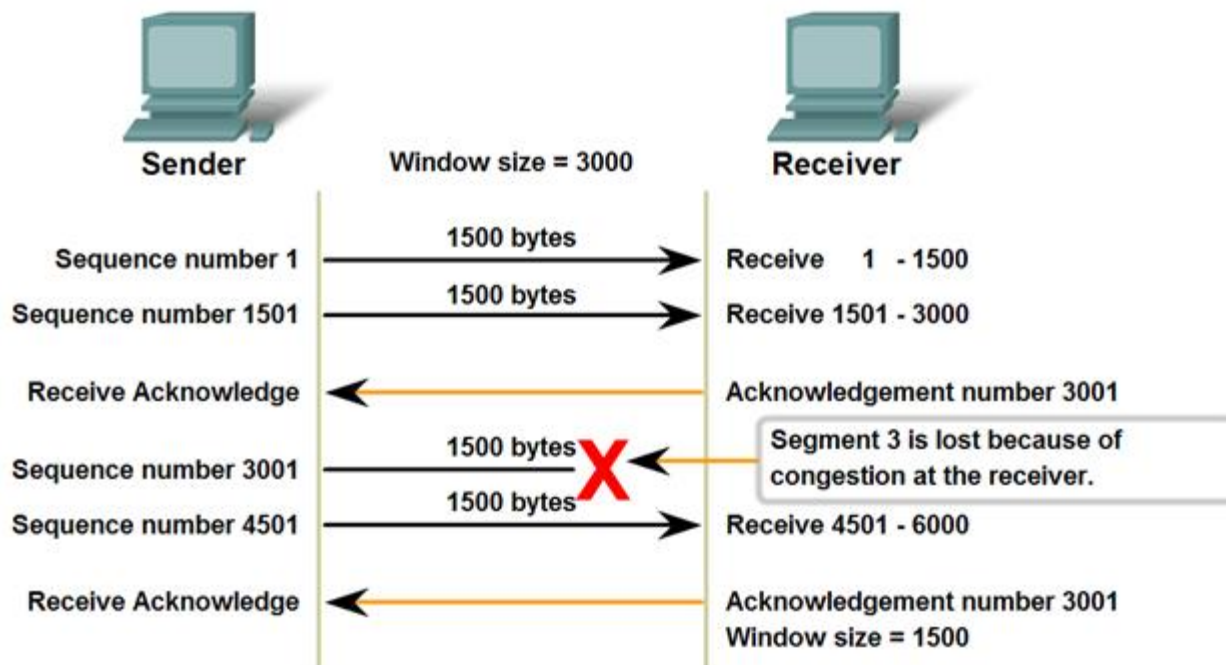
◆ PDU=segment



Transport Layer Functions

◆ Transmissions and Retransmissions

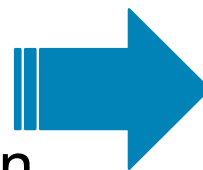
TCP Congestion and Flow Control



If segments are lost because of congestion, the Receiver will acknowledge the last received sequential segment and reply with a reduced window size.

TCP–Transmission Control Protocol

- Connection oriented
- Transmission/Retransmission



Reliable

UDP



IP Telephony



Streaming Live Video

Required protocol properties:

- Fast
- Low overhead
- Does not require acknowledgements
- Does not resend lost data
- Delivers data as it arrives

TCP



SMTP/POP (Email)



HTTP

Required protocol properties:

- Reliable
- Acknowledge data
- Resends lost data
- Delivers data in order sent

UDP – User Datagram Protocol

- Connection-Less
 - Transmission/Retransmission
- Unreliable

UDP



IP Telephony



Streaming Live Video

Required protocol properties:

- Fast
- Low overhead
- Does not require acknowledgements
- Does not resend lost data
- Delivers data as it arrives

TCP



SMTP/POP (Email)



HTTP

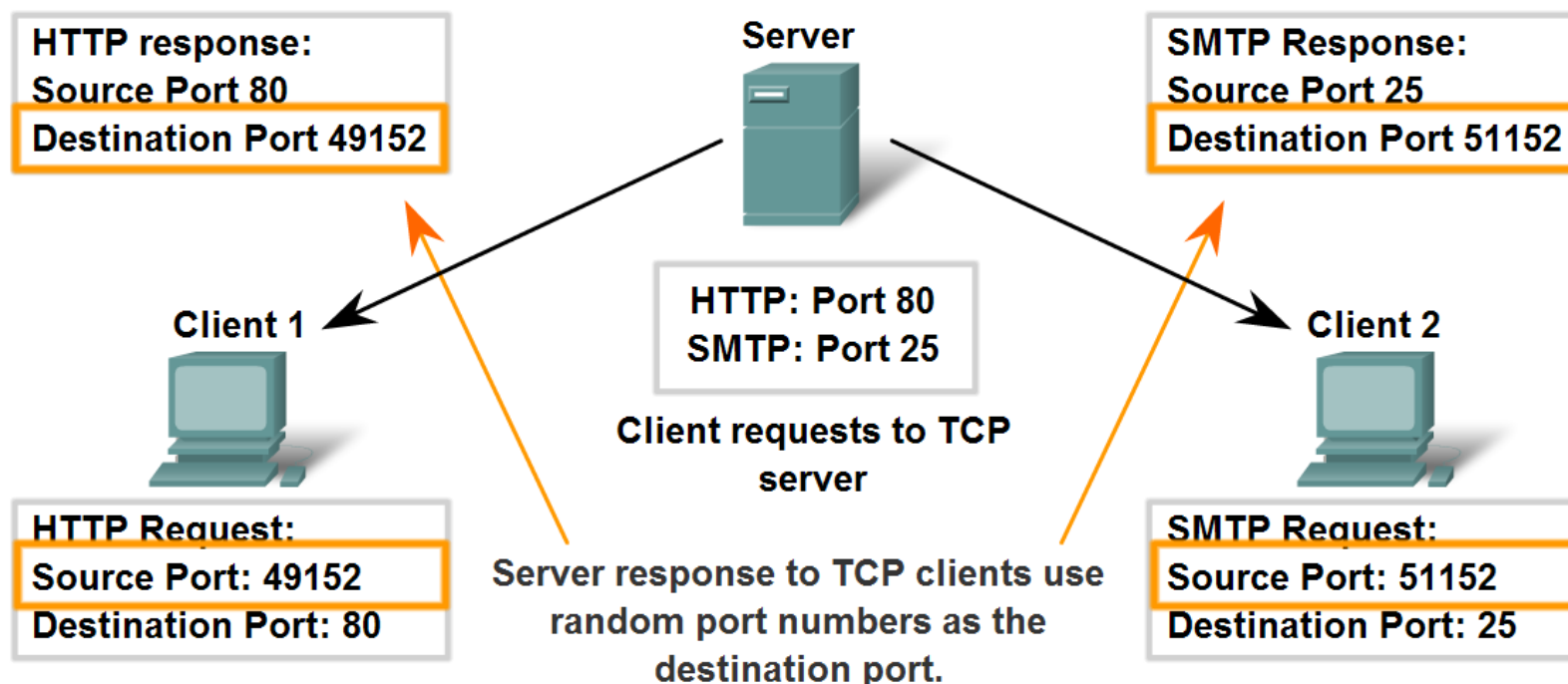
Required protocol properties:

- Reliable
- Acknowledge data
- Resends lost data
- Delivers data in order sent

Port Numbers

- Identify Source and Destination Apps Communicating

Clients Sending TCP Requests



Port Numbers

- Connection : Source Socket– Destination Socket
- NETSTAT – win/linux/unix command

192.168.1.200:53001 - 83.156.12.13:80

Port Numbers

- Connection : Source Socket– Destination Scket
- NETSTAT – win/linux/unix command

```
C:\>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	kenpc:3126	192.168.0.2:netbios-ssn	ESTABLISHED
TCP	kenpc:3158	207.138.126.152:http	ESTABLISHED
TCP	kenpc:3159	207.138.126.169:http	ESTABLISHED
TCP	kenpc:3160	207.138.126.169:http	ESTABLISHED
TCP	kenpc:3161	sc.msn.com:http	ESTABLISHED
TCP	kenpc:3166	www.cisco.com:http	ESTABLISHED

Port Numbers

Well Known PN: 0 – 1023

TCP

20,21 FTP
 22 SSH
 23 Telnet
 25 SMTP
 110 POP3
 143 IMAP
 80 HTTP
 443 HTTPs

UDP

67,68 DHCP
 69 TFTP
 123 NTP
 520 RIP

TCP&UDP

53 DNS
 118& 156 SQL Services

Port Numbers

Reserved PN: 1024 – 49151

TCP

3650 – 3665 Skype
1645,1646 TACACS+
8080 HTTP alternate
8008 HTTP alternate

UDP

1812,1813 RADIUS
3724 WOW
5004 RTP

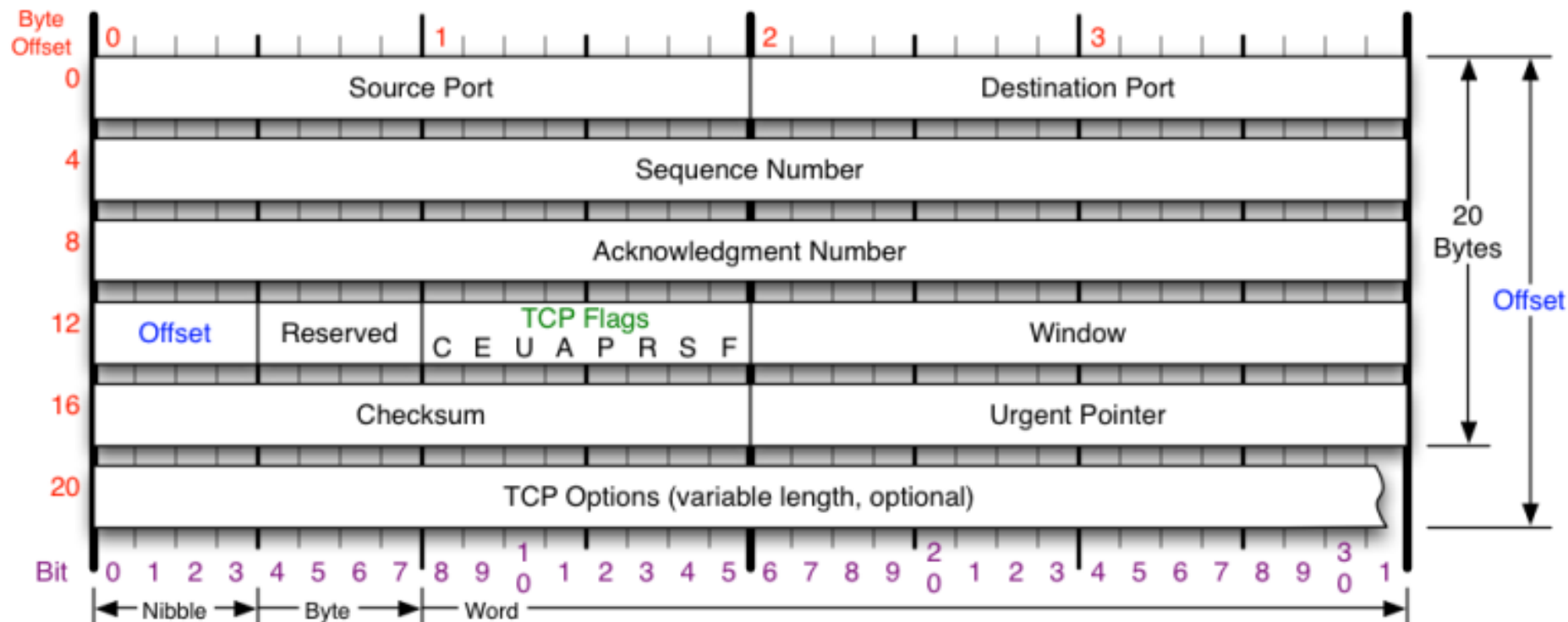
TCP&UDP

2000 Cisco SCCP
5060 SIP (UDP)
1433 MySQL
2948 WAP (MMS)

Dynamic Allocated Port: 49152 - 65535

TCP Header

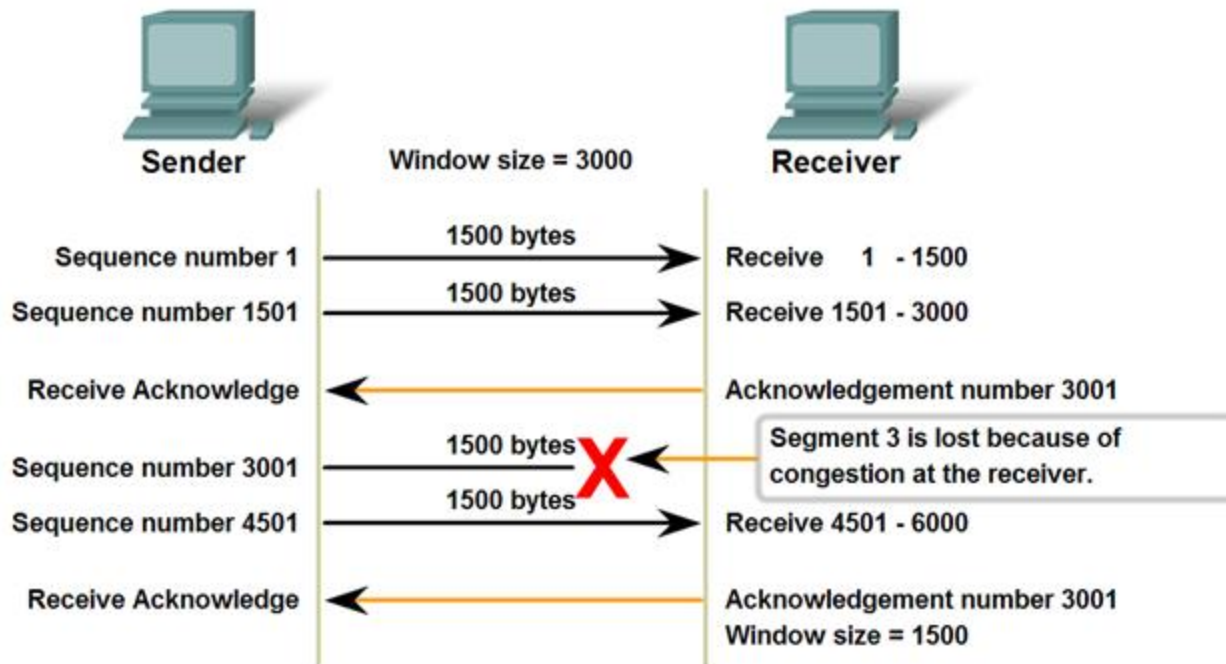
TCP Segment



TCP Header

Sequence & Acknowledgement Numbers

TCP Congestion and Flow Control



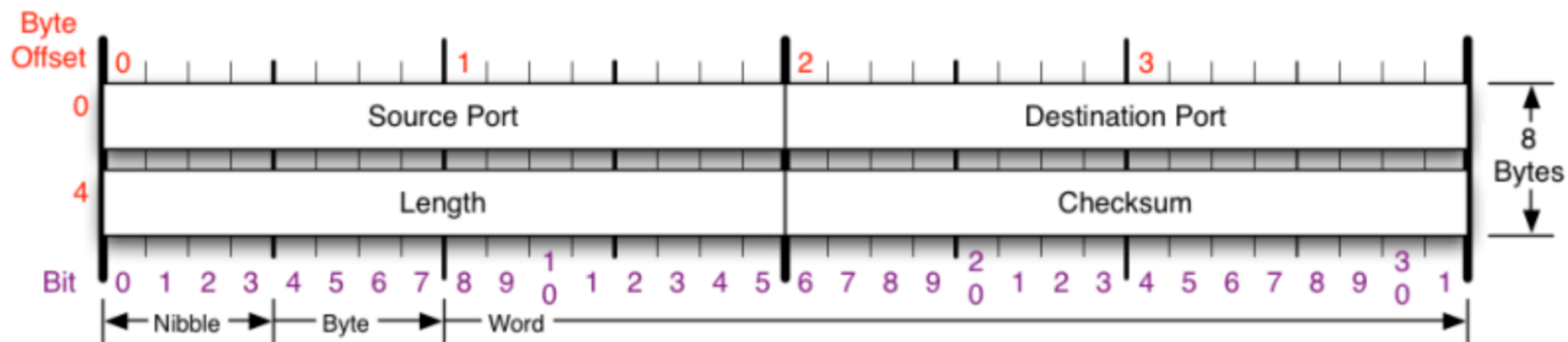
If segments are lost because of congestion, the Receiver will acknowledge the last received sequential segment and reply with a reduced window size.

TCP Header

- Type of TCP Segment:
 - Urgent
 - Acknowledgement
 - Push
 - Reset
 - Synchronize
 - Finalize

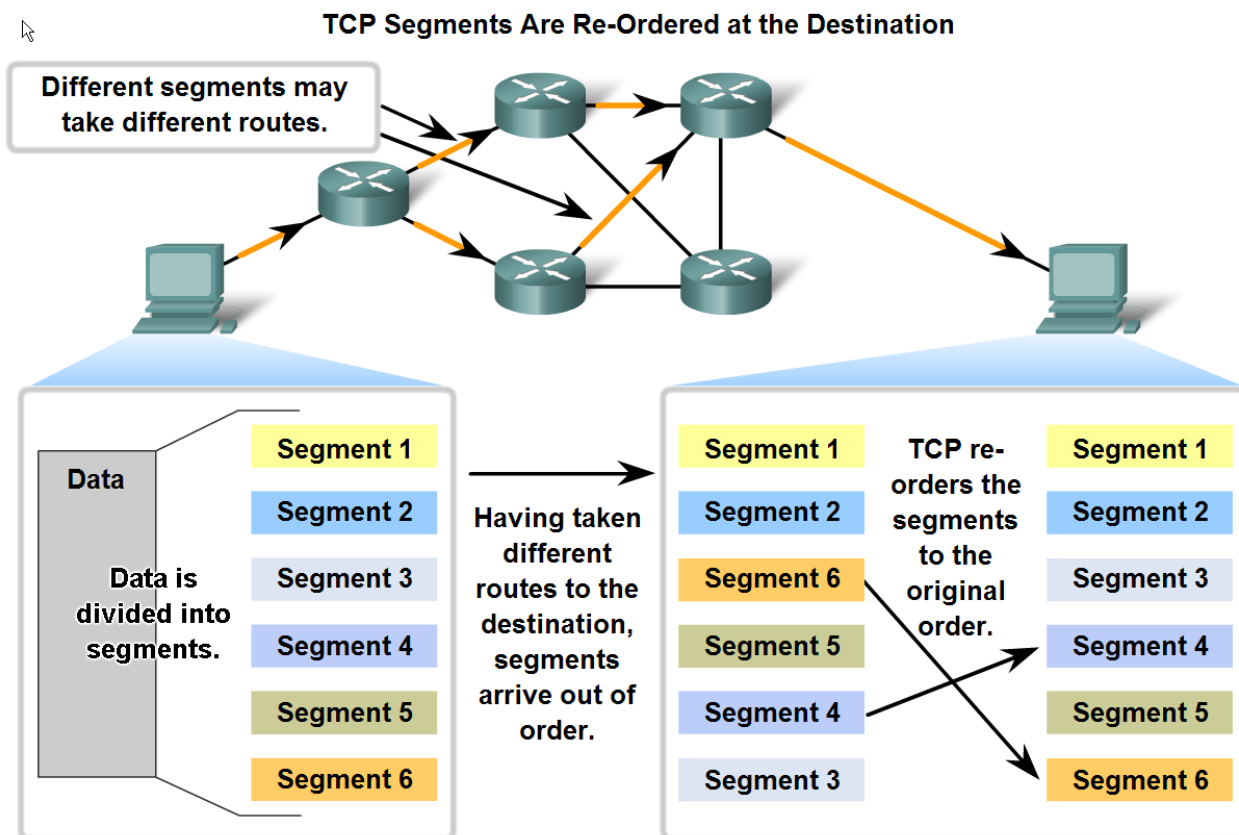
UDP Header

- UDP Segment = Datagram



TCP Connection

■ Data Flow



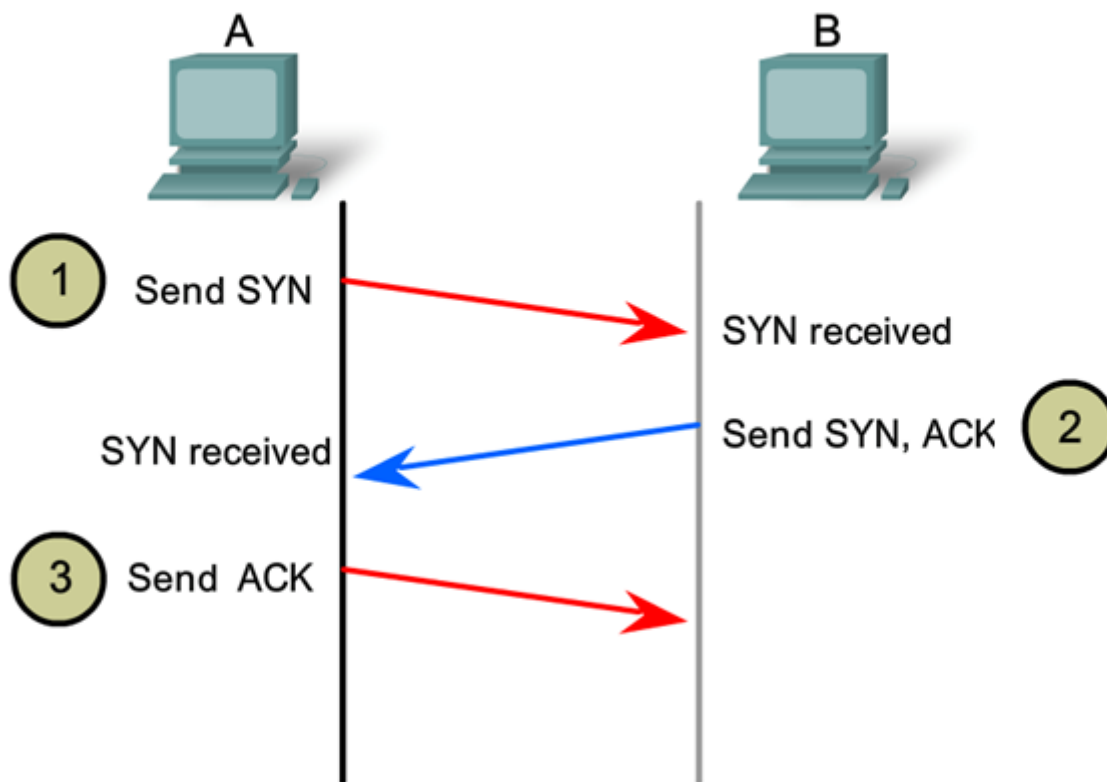
TCP Connection

- Start Connection
- Data/Protocol transfer
- End Connection

TCP Connection

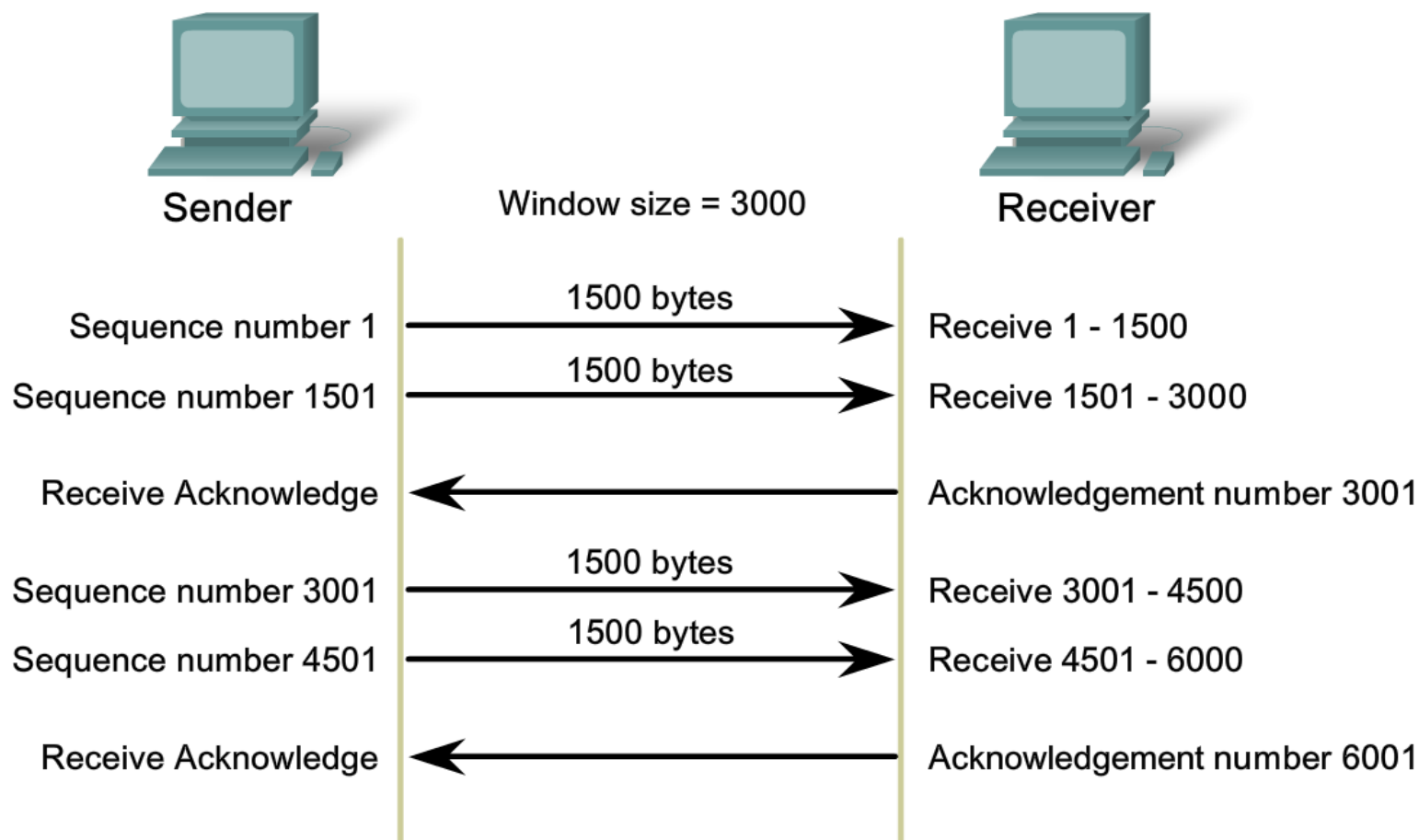
- Connection Establishment= The Three Way Hand-Shake

TCP Connection Establishment and Termination



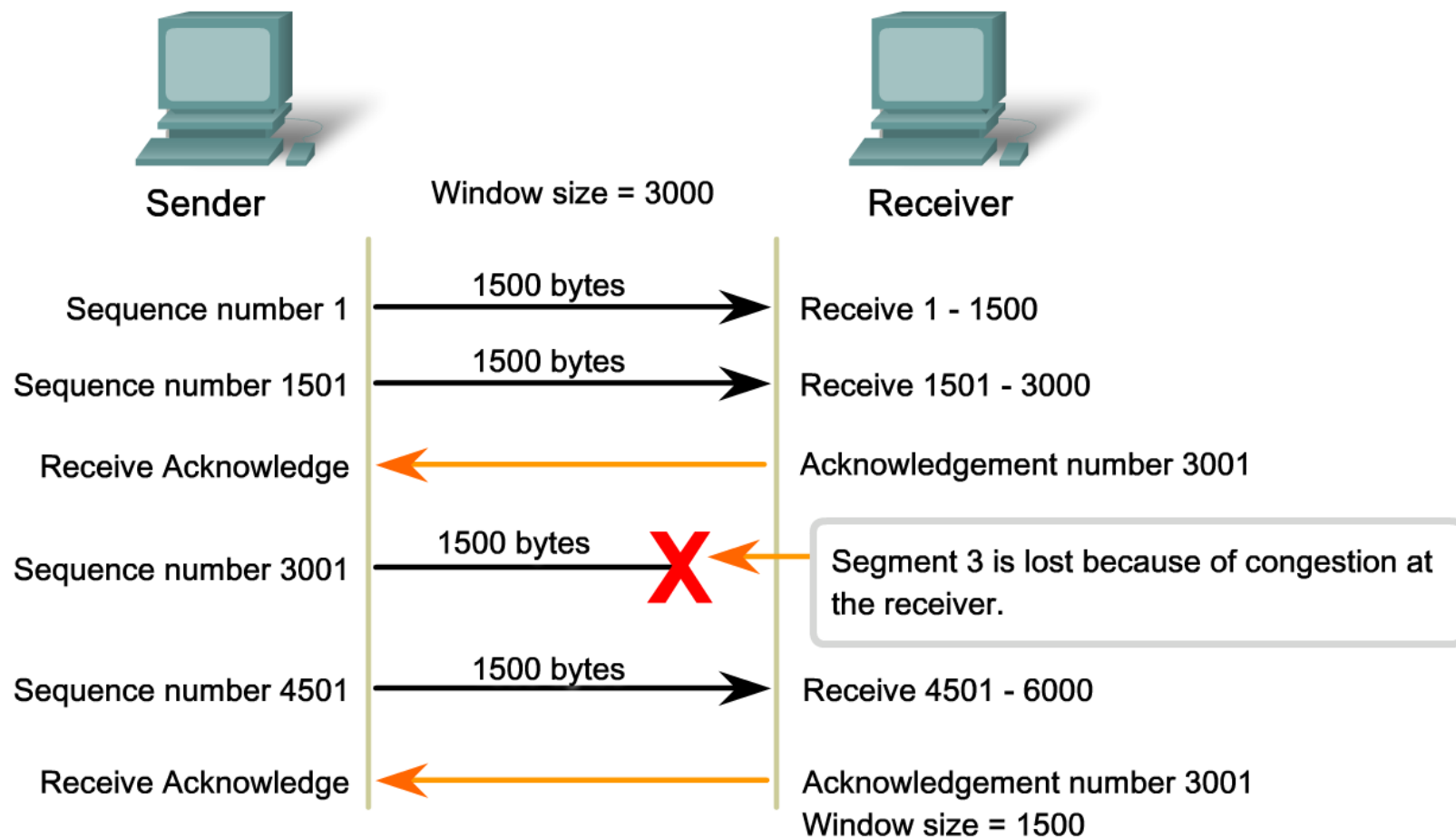
TCP Connection

■ Data Flow Control



TCP Connection

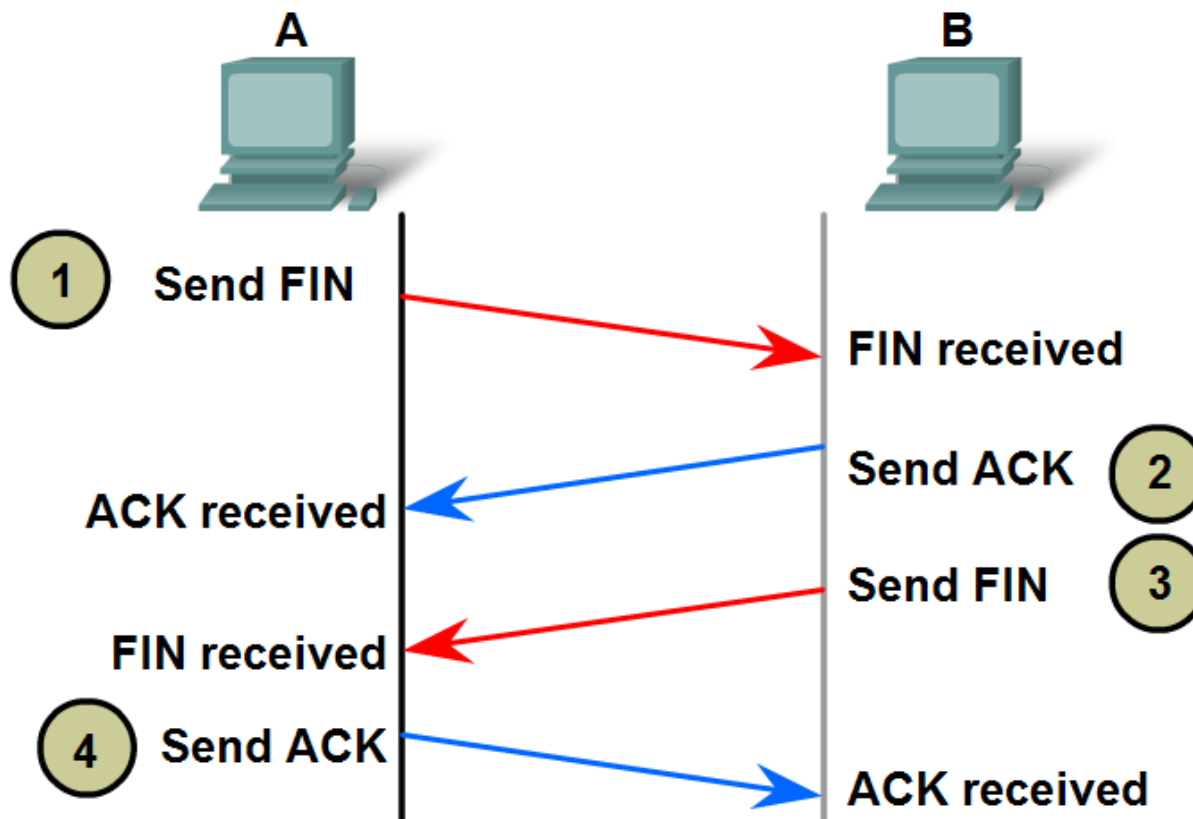
▪ Data Flow Control



TCP Connection

■ Connection Termination

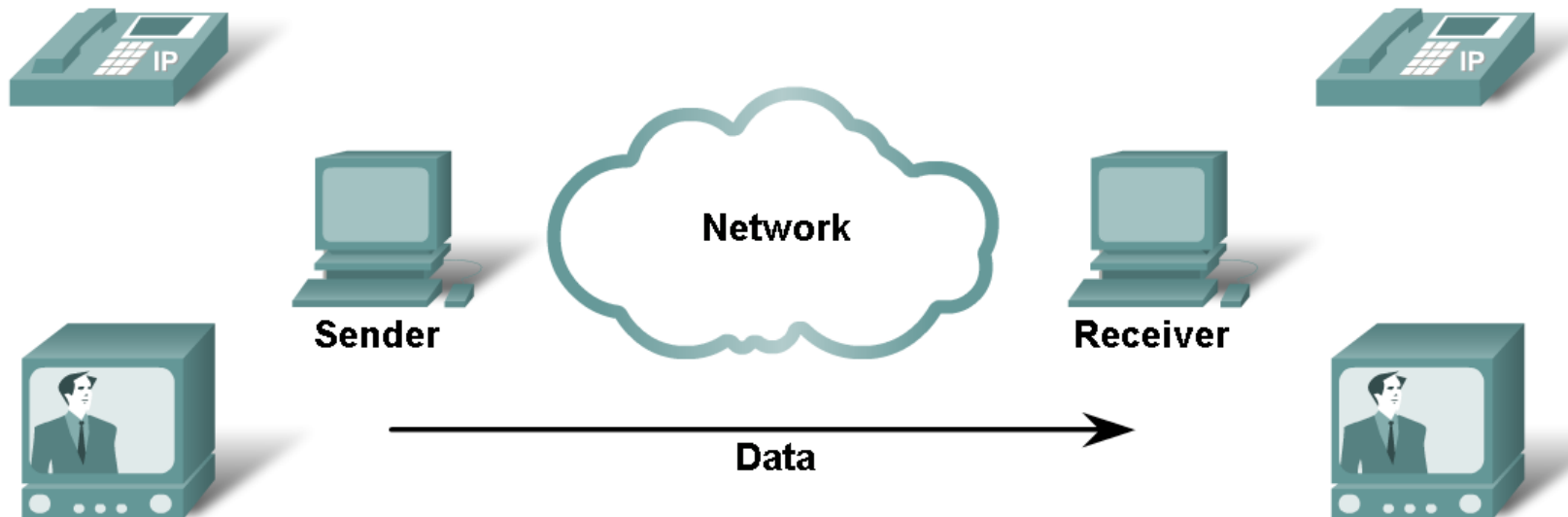
🖱️ TCP Connection Establishment and Termination



UDP

- Connection-Less -» Unreliable
- No Overhead Needed

UDP Low Overhead Data Transport



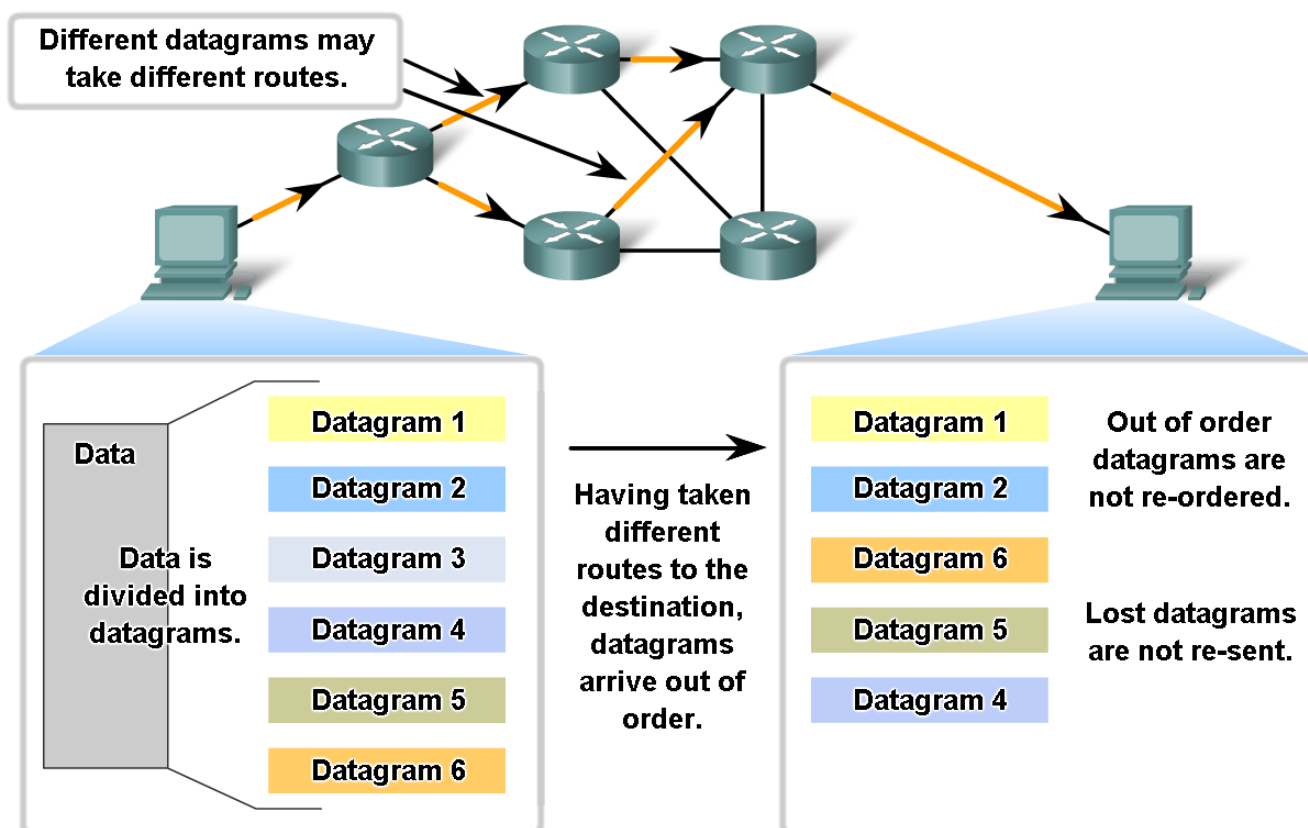
UDP does not establish a connection before sending data.

UDP

■ Best-Effort Delivery protocol

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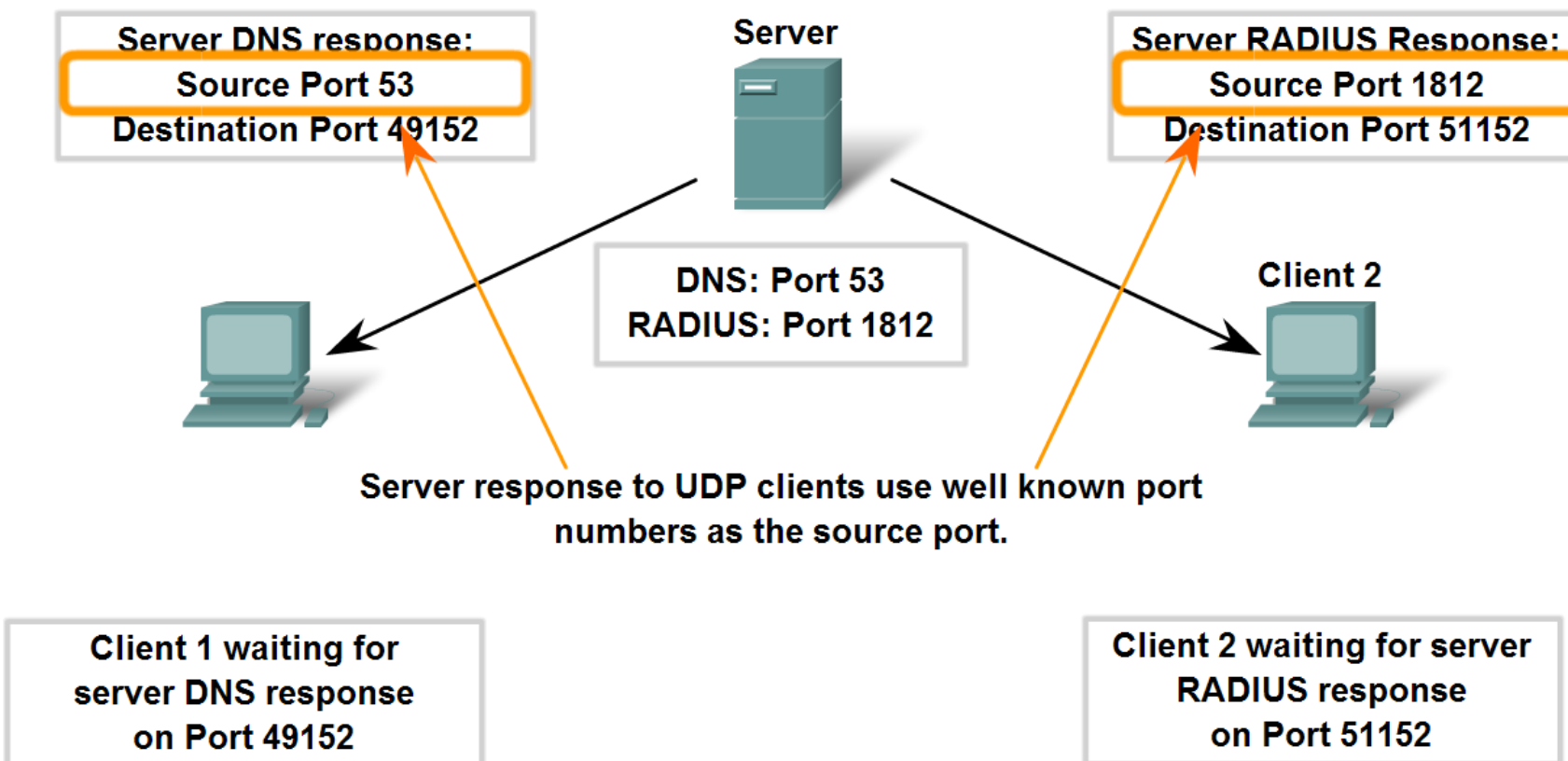
UDP: Connectionless and Unreliable



UDP

Example of real world UDP Operation: DNS


Clients Sending UDP Requests



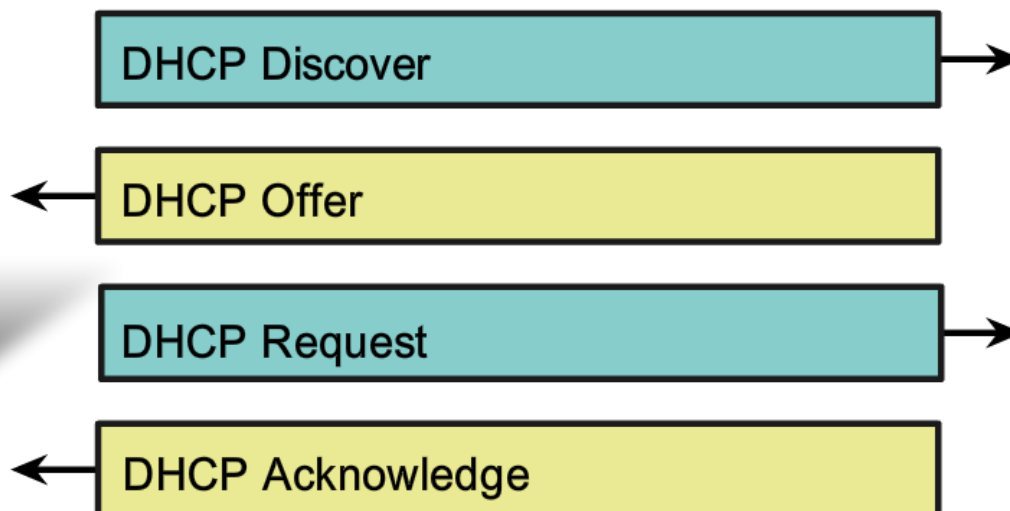
UDP

Example of real world UDP Operation: DHCP

DHCP Client



DHCP Server



Summary

- Transport Layer encapsulation
- Role of TCP and UDP
- Connections managed by TCP
- When reliability wanted TCP is used
- When overhead not wanted UDP is used
- LAB: Wireshark Capture

